

Use-Case Flow Specification

Use Case Name: Reserve Charging Slot

Primary Actor: EV Driver

1. Preconditions

1. The EV Driver has accessed the EV Charging Station Reservation system
2. The driver's vehicle connector type is known.
3. A compatible charging station is available.
4. At least one charging socket and time slot is available.

2. Postconditions

1. The charging slot is successfully reserved.
2. The selected socket and time slot are no longer available to other drivers.
3. A unique reservation ID is generated.
4. The charging cost is calculated using the applicable dynamic power rate.

3. Main Success Scenario

1. The EV Driver opens the system.
2. The driver searches for nearby charging stations.
3. The system displays compatible charging stations.
4. The driver selects a charging station.
5. The system displays available charging sockets and time slots.
6. The driver selects an available 30-minute charging slot.
7. The system checks the vehicle connector compatibility.
8. The system calculates the charging cost using the applicable dynamic power rate
9. The driver confirms the reservation.
10. The system generates a unique reservation ID.
11. The system marks the selected socket and time slot as reserved.
12. The system displays the reservation confirmation.

4. Alternate Flow — Driver Does Not Plug In

1. The driver successfully reserves a charging slot.
2. The driver does not plug the vehicle into the charger within 10 minutes of the slot start.
3. The system detects that the driver has not plugged in.
4. The system releases the locked charging socket.
5. The system applies the applicable penalty.
6. The charging socket becomes available for another reservation.